

Alexander Chang

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EDUCATION

University of California, Berkeley

GPA: 3.973/4.0

Bachelor's Degree in Computer Science, Data Science, and Cognition

May 2027

- **Relevant Coursework:** Data Structures and Algorithms, Intractable Problems, Machine Learning, Data Engineering, Data Mining, Principles and Techniques of Data Science, Computer Programs, Computer Architecture, Computational Cognitive Science, Probability Theory, Discrete Math, Linear Algebra, Differential Equations.
- **Clubs/Activities:** UpSync Technology Consulting, Computer Science Mentors (Data Structures and Algorithms Mentor), UC Berkeley Symphony Orchestra (Associate Principal Cellist), Basketball, Soccer, Golf, Cooking.

EXPERIENCE

Amazon

Jun 2025 – Aug 2025

Software Development Engineer Intern

Seattle, WA

- Designed and implemented a production RAG pipeline to automate responses to AWS account security tickets. Leveraged AWS Bedrock Knowledge Base, Lambda, and deployed via an AWS CDK-based CI/CD pipeline.
- Integrated ticket scraping, classification, and filtering logic to process 1,000+ live tickets daily, instantly compiling responses and reducing average response time by up to 1,500× while significantly reducing manual workload.
- Utilized OpenSearch vector search for improved accuracy and AWS Step Functions to handle failures and retries.

TotalEnergies

Jan 2025 – May 2025

Machine Learning Project Manager – Contract

Berkeley, CA

- Led a team of 7 engineers to predict electricity prices using SARIMA, LSTMs, and ensemble deep learning models.
- Built and trained models using historical and real-time data collected with a scheduled Python web scraper.
- Utilized feature forecasting, axis shifting, and hyperparameter tuning, achieving R2 values between 0.91–0.97.

AMD

Aug 2024 – Jan 2025

Software Engineer – Contract

Berkeley, CA

- Designed and implemented a full-stack dashboard, enabling efficient management and visualization of 10,000+ metrics, documents, and images. AMD is using the dashboard in court to handle product compliance court cases.
- Leveraged Microsoft SQL Server and Azure Blob Storage to organize structured and unstructured data.
- Developed an API in Python using Azure Functions to facilitate dynamic SQL querying, document storage, retrievals, and uploads to designated containers. Built the front-end using React.js and Axios.

PROJECTS

Computer Science Mentors Website Waitlist Feature

- Designed and implemented a waitlist system for the Berkeley Computer Science Mentors website, serving 1,000+ students. Used Python, Django, RESTful APIs, and a PostgreSQL database to implement the backend.
- Integrated PostgreSQL queries with the API, enabling real-time data retrieval for the JavaScript frontend.

Email Spam Detector

- Built a logistic regression model in Python to classify emails as spam or legitimate with over 90% accuracy.
- Utilized Scikit-learn to build/train the model. Investigated keywords and structural email features for advanced feature engineering, incorporating cross-validation to optimize accuracy, precision, and recall.
- Developed the project in Jupyter Notebook, using Seaborn and Matplotlib for exploratory data analysis.

Multi-Circle Trochoid Animation

- Designed and implemented an animation website tool using HTML and JavaScript. This tool draws the curve that is traced by following a sequence of up to 5 circles rolling along the inside or outside of each other's circumferences.
- Derived the mathematical formula using algebra and trigonometry, and translated the logic into JavaScript code.

SKILLS

- **Software:** Python, Java, C, Pandas, NumPy, Scikit-learn, Flask, React, AWS, Azure, SQL, JavaScript, HTML, CSS, Git, VS Code, IntelliJ, Jupyter Notebook, Microsoft Office. **Languages:** English, Mandarin, Spanish